Inverter Box Setup Guide

In this document, we will explain how to control the inverter switches on the Inverter box via your laptop. Here is the path:

Laptop -> Launchpad -> Interface Board -> Inverter Box

Equipment Required:

Item	Role
1 x Laptop	For flashing firmware onto the LaunchPad and communicating with it while it runs
1 x SEMITEACH – IGBT from SEMIKRON	The inverter box: 3-phase IGBT inverter with rectifier and brake chopper (the power stage)
1 x LAUNCHXL-F28P55X or an equivalent	The control board: its MCU generates the PWM and reads the measurements
1 x Interface Board	Connects the LaunchPad to the inverter box: gate-drive outputs, error inputs, thermal-trip interlock, and probe signal conditioning
1 x HP E3631A	Bench supply providing the +15 V auxiliary power for the inverter box and the interface board (use the +25 V output set to +15 V, current limit ~1 A)
1 x USB-C to C or USB-C to A cable	Connects the laptop to the LaunchPad (flashing + real-time communication); must be a data cable, not charge-only

Setup Procedure

Safety: make every connection with all supplies off. Apply the +15 V control power and check the status LEDs before energizing the power stage. Never re-wire while the DC bus is energized. After switching off, the DC bus stays charged for tens of seconds (internal balancing resistors, $\tau \approx 46$ s). Before re-wiring anything, measure the bus and confirm it is below 10 V (5 V in the final project; see its handout).

1. Plug the LaunchPad into sockets J1-J4 on the interface board, USB connector toward the DE-9 connector edge.

2. On the LaunchPad, make sure switch S4 is UP (its default). If it is down, two of the Inverter 2 gate outputs (Top 3 and Bot 3) will not work: S4 down routes GPIO4/GPIO5 to the on-board CAN transceiver.

3. Connect the thermal-trip leads from the inverter box to the Thermal Trip 1 jacks (Thermal Trip 2 for a second inverter). An inverter's gate-drive outputs stay disabled until its trip loop is closed.

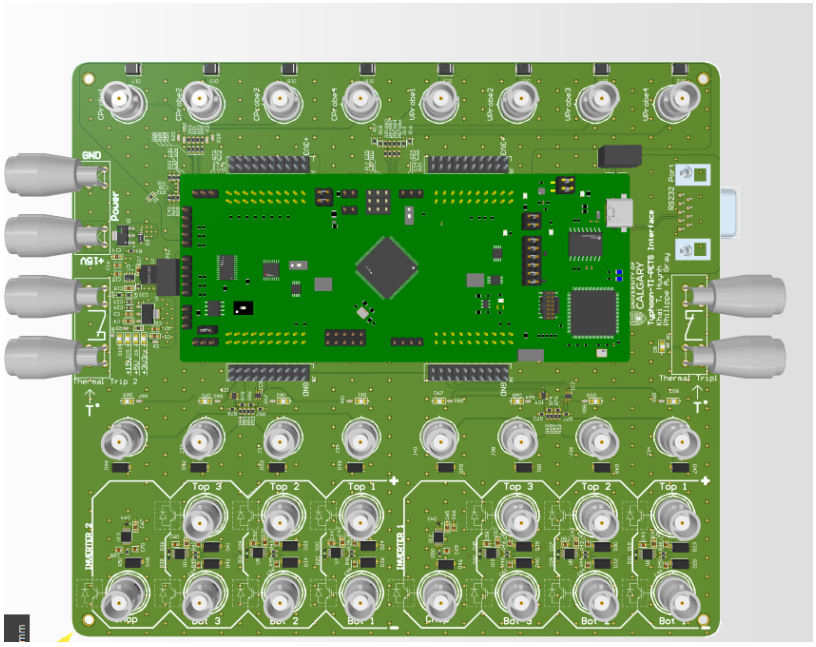
4. Connect the gate-drive BNCs to the inverter gate inputs and the inverter error outputs to the Error BNCs - the labels match on both ends (Top 1 to Top 1, and so on). The error signals are not monitored this year (see the Configuration Table), but connect them anyway. Connect voltage/current probes to the VProbe/CProbe inputs as the lab requires.

5. Connect +15 V and GND to the power banana jacks from the HP E3631A: use the +25 V output, set to +15 V with a ~1 A current limit, and turn the output on.

6. Check the LEDs: D5, D9 and D4 (supply rails) on; D8/D10 on when the trip loops are closed.

7. Connect the laptop to the LaunchPad over USB and flash the firmware. JP1 on the LaunchPad stays open: USB carries data only, and the LaunchPad is powered from the interface board, so the +15 V supply must be on before you flash.

Interface Board



High Level Summary

The interface board is designed to control up to 2 x three-phase inverter boxes. The interface board supports up to four voltage inputs and four current inputs. As shown in the figure above, a TI LaunchPad can be plugged directly into the board.

Configuration Table

Description	Details
PWM Outputs	14 gate-drive outputs: Top/Bot for each of three legs plus a chopper drive, for up to two inverters
Digital Inputs (Dedicated for Error)	8 error inputs (not monitored this year; connect the Error BNCs anyway). Fault supervision is provided by the thermal-trip interlock and by the gate drivers' own built-in protection inside the inverter box.
Analog Inputs	4 voltage-probe and 4 current-probe BNC inputs
Power Input	+15 V and GND from the auxiliary bench supply, via the power banana jacks

Protection Related

Thermal Trip 1	Connect from the inverter box to the interface board. The Inverter 1 PWM outputs are enabled only while this loop is closed; the inverter opens it above 71 °C.
Thermal Trip 2	Same as Thermal Trip 1, for the Inverter 2 PWM outputs.

Status Check

+15 V rail	LED D5 is ON when +15 V is reaching the board
+5 V rail	LED D9 is ON when the on-board 5 V rail is up
+3.3 V rail	LED D4 is ON when the on-board 3.3 V rail is up
Thermal trip	LEDs D8/D10 are ON when the corresponding trip loop is closed (no over-temperature). OFF if the loop is disconnected or the module has exceeded 71 °C.
Error LEDs	Non-functional on this board revision — they stay lit regardless of the error state. Ignore them.

Sensing Circuits

Each probe input is conditioned before it reaches the MCU: BNC input (±7 V max) → resistor divider → active low-pass filter (built around an OPA4197) → ADC pin. The output is centred at about 1.62 V, so 0 V at the BNC reads as about 1.62 V at the ADC. This is the measured bench value; earlier versions of this guide gave 1.667 V. Check your own board's zero reading with the probe connected and nothing flowing. The Typhoon HIL project models use the same 1.62 V offset and gains. Choose the probe setting so the signal stays within ±7 V at the BNC.

DC scaling:

Voltage channels: V_ADC = 1.62 V + 0.213 × V_BNC. On the 10:1 probe setting this is ≈21.3 mV per volt, so ±70 V spans ≈0.13 V to 3.11 V at the ADC. The ADC saturates at 3.3 V, i.e. at

about +79 V (and about −76 V) on the 10:1 setting: above that the reading stops rising, so set any DC-voltage trip below it.

Current channels: V_ADC = 1.62 V + 0.154 × V_BNC. On the 100 mV/A probe setting this is ≈15.4 mV per amp. The 1146B's output never exceeds about ±1 V (10 A peak on the 100 mV/A range, 100 A on 10 mV/A), so a current channel uses only a small part of the ADC range: ±10 A at 100 mV/A spans ≈1.47 V to 1.77 V.

Dynamics: each channel behaves as a low-pass filter with two poles near 700–900 Hz and a zero near 1.5 kHz. In practice: 60 Hz passes with the DC gain given above (within about 0.5 %), but the gain is about 5 % low at 200 Hz and about 25 % low at 500 Hz; switching-frequency content is strongly attenuated (about −21 dB at 5 kHz and −27 dB at 10 kHz, falling −20 dB/decade beyond). The filter contributes noticeable phase lag from a few hundred hertz — about 20° at 200 Hz — which must be included in the loop model when designing closed-loop controls.

Pin Mapping

The tables below show, for each connector on the interface board, the LaunchPad pin and the MCU signal behind it — use these when configuring the firmware. “Interface board header” is the matching pin on the board’s J5-J8 headers.

Table 1 - Gate-drive PWM outputs (BNC to inverter)

Inverter	Output	Driver net	LaunchPad pin	Device signal	Interface board header
1	Top 1	IN_Top_1	J4.40	GPIO0 / EPWM1_A	J6.1
1	Bot 1	IN_Bot_1	J4.39	GPIO1 / EPWM1_B	J6.3
1	Top 2	IN_Top_2	J4.38	GPIO2 / EPWM2_A	J6.5
1	Bot 2	IN_Bot_2	J4.37	GPIO3 / EPWM2_B	J6.7
1	Top 3	IN_Top_3	J4.36	GPIO10 / EPWM6_A	J6.9
1	Bot 3	IN_Bot_3	J4.35	GPIO11 / EPWM6_B	J6.11
1	Chop	IN_Chop	J8.80	GPIO12 / EPWM7_A	J8.1
2	Top 1	IN2_Top_1	J8.78	GPIO6 / EPWM4_A	J8.5

2	Bot 1	IN2_Bot_1	J8.77	GPIO7 / EPWM4_B	J8.7
2	Top 2	IN2_Top_2	J8.73	GPIO63 / EPWM10_B	J8.15
2	Bot 2	IN2_Bot_2	J8.74	GPIO62 / EPWM10_A	J8.13
2	Top 3	IN2_Top_3	J4.31	GPIO5 / EPWM3_B	J8.11
2	Bot 3	IN2_Bot_3	J4.32	GPIO4 / EPWM3_A	J8.9
2	Chop	IN2_Chop	J8.79	GPIO13 / EPWM7_B	J8.3

Note, each leg's Top/Bot pair is the A/B pair of one ePWM module, so complementary PWM with dead-band can be generated per leg. On Inverter 2, legs 2 and 3 have Top on the B output (EPWM10_B, EPWM3_B), the reverse of every other leg; configure those modules accordingly.

Table 2 - Analog probe inputs

Input	Net	LaunchPad pin	ADC channels (same pin)	Interface board header
VProbe 1	VProbe1MCU	J3.24	ADCINB2 / ADCINC6	J5.8
VProbe 2	VProbe2MCU	J3.23	ADCINA6	J5.6
VProbe 3	VProbe3MCU	J3.25	ADCINA11 / B10 / C0	J5.10
VProbe 4	VProbe4MCU	J3.26	ADCINA14 / B14 / C4	J5.12
CProbe 1	CProbe1MCU	J7.64	ADCINA10 / B1 / C10	J7.8
CProbe 2	CProbe2MCU	J7.63	ADCINB0 / C11	J7.6
CProbe 3	CProbe3MCU	J7.65	ADCINA5	J7.10
CProbe 4	CProbe4MCU	J7.66	ADCINA4 / B8	J7.12